

claude-windows / 42211f80-2169-4025-ad9d-99c1aa29ff70

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\42211f80-2169-4025-ad9d-99c1aa29ff70\subagents\workflows\wf_6237b110-d58\agent-a6056ad5d1719f198.jsonl`  
- SHA-256: `f198a0bdb6da264ecfdec58635f8bd1a3d32afe54a2e892c450a9adf33833d69`  
- Source modified: `2026-06-06T17:35:16+00:00`  
- Imported at: `2026-07-05T16:48:28+00:00`  
- project: `wf_6237b110-d58`  
- session_id: `42211f80-2169-4025-ad9d-99c1aa29ff70`
```

Transcript

1. user (2026-06-06T17:33:43.099Z)

Adversarial Claim Verifier (voter 3/3)

Be SKEPTICAL. Try to REFUTE this claim. ?2/3 refutations kill it.

Research question

Research alternative methods to automatically detect rally start/end segments (temporal segmentation) in racket-sport video ? especially badminton ? for a SELF-HOSTED, LOCAL-FIRST pipeline, and assess whether a cloud-VLM teacher ? local-model distillation is the best approach or whether other methods are competitive or complementary. Keep it focused ("mini" research): prioritize actionable, commercially-licensable, locally-runnable approaches over exotic SOTA needing huge compute or restrictive licenses.

OUR CONTEXT / CONSTRAINTS (ground every recommendation in these):

- Self-hosted on a single consumer GPU (RTX 2070 Super), Windows + WSL2. We want LOCAL-ONLY inference at runtime ? privacy lever: user video never leaves the device. It's a commercial product, so PERMISSIVE / commercial-friendly licenses are REQUIRED (avoid AGPL and non-commercial-only weights/datasets).
- Current substrate: WASB (MIT) shuttle-trajectory detector ? per-frame (x, y, visibility). We derive rally windows via velocity/merge-gap/min-duration thresholds plus a net-crossing gate. We ALSO have a permissive person detector available (torchvision Faster R-CNN + supervision ByteTrack) and the video's AUDIO track.
- Current best approach: use paid Gemini (cloud VLM, full-context) OFFLINE as a TEACHER to label rallies, then DISTILL into a local model (logistic regression on shuttle-trajectory features). Findings so far: threshold-tuning ceilings around F1 ? 0.44 and overfits with few videos; the learned model is data-starved at n=2 labeled videos; WASB candidate RECALL is a real bottleneck (its windows sometimes don't align with true rallies at IoU 0.5).
- Inputs we can exploit locally: video frames, the shuttle trajectory, player bounding boxes / pose (if we add a permissive pose model), and the AUDIO track (shuttle "thwack" hits, crowd noise, umpire/score calls).

SPECIFICALLY COVER:

1. Method families for rally/point segmentation in racket sports (badminton, tennis, table tennis, pickleball): audio-based shot/hit detection; temporal action localization (TAL) deep models; pose/player-stance + court-homography state machines (e.g., "serve-ready stance" gating); classical change-point detection / HMM on the shuttle trajectory; scoreboard / score-overlay OCR for point boundaries; self-supervised / weakly-supervised approaches; and VLM teacher?student distillation.
2. For EACH family: how it works, the labels/data it needs, expected accuracy, local-inference COMPUTE cost, and LICENSE / commercial viability of any models, datasets, or weights involved.
3. Badminton rally-segmentation datasets (e.g., ShuttleSet, TrackNet/TrackNetV2 data) and their licenses / commercial usability.
4. A RANKED recommendation for our local-first, permissive, single-GPU setup: is Gemini-teacher ? local-distillation the best, or are some alternatives competitive or complementary ?

especially AUDIO-based hit detection and pose/court-geometry state machines, which we can run fully locally?

5. Concrete, cheap next experiments we could run with what we already have (shuttle trajectory + person detector + audio).

Claim under review

"This rally segmentation method imposes the same heuristic temporal constraints the user's pipeline uses (minimum rally duration and minimum inter-rally gap): a rally is at least 3 seconds long with a minimum 3-second gap between rallies."

****Source:**** <https://arxiv.org/pdf/2308.12199> (primary)

****Supporting quote:**** "A rally is at least 3 seconds long and there is a minimum of 3 seconds gap between two rallies."

Checklist

1. Is the claim actually supported by the quote, or is it an overreach/misread?
2. WebSearch for contradicting evidence ? does any credible source dispute or heavily qualify this?
3. Is the source quality sufficient for the claim's strength? (extraordinary claims need primary sources)
4. Is the claim outdated? (check dates ? old claims about fast-moving fields are suspect)
5. Is this a marketing claim / press release / cherry-picked benchmark / forum speculation?

****refuted=true**** if: unsupported by quote / contradicted / low-quality source for strong claim / outdated / marketing fluff.

****refuted=false**** ONLY if: claim is well-supported, current, and source quality matches claim strength.

Default to refuted=true if uncertain.

Structured output only. Evidence MUST be specific.

2. assistant (2026-06-06T17:33:46.287Z)

I'll verify this claim about the 3-second rally duration and gap constraints. Let me start by fetching the primary source.

3. assistant (2026-06-06T17:33:51.362Z)

I'll fetch the primary source paper to verify the exact quote and context.

4. user (2026-06-06T17:33:54.170Z)

maxContentLength size of 10485760 exceeded

5. assistant (2026-06-06T17:33:58.211Z)

The PDF is too large to fetch directly. Let me try the arxiv abstract page and search for the paper details.

6. user (2026-06-06T17:34:03.161Z)

Paper Summary

****Title:**** Towards Real-Time Analysis of Broadcast Badminton Videos

****Authors:**** Nitin Nilesh, Tushar Sharma, Anurag Ghosh, and C. V. Jawahar

****Sport:**** This paper focuses specifically on ****badminton****, not tennis or other racket sports.

****Rally/Temporal Segmentation:**** Yes, the work does address this aspect. The authors note they "focus on the gameplay by removing replays and other redundant parts of the broadcast match."

****Abstract Quote:****

"Analysis of player movements is a crucial subset of sports analysis...we propose an end-to-end framework for player movement analysis for badminton matches on live broadcast match videos."

The framework calculates distance covered by players, tracks their movements, and generates heatmaps of court coverage using only visual input from broadcast feeds.

7. user (2026-06-06T17:34:05.018Z)

Web search results for query: "arxiv 2308.12199 "rally is at least 3 seconds" badminton segmentation"

Links: [{"title": "[2308.12199] Towards Real-Time Analysis of Broadcast Badminton Videos", "url": "https://arxiv.org/abs/2308.12199"}, {"title": "Towards Real-Time Analysis of Broadcast Badminton Videos", "url": "https://arxiv.org/pdf/2308.12199"}, {"title": "MonoTrack: Shuttle trajectory reconstruction from monocular badminton video", "url": "https://arxiv.org/pdf/2204.01899"}, {"title": "Long rallies and next rally performances in elite men's and women's badminton", "url": "https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7053721/"}, {"title": "TennisExpert: Towards Expert-Level Analytical Sports Video Understanding", "url": "https://arxiv.org/pdf/2603.13397"}, {"title": "BST: Badminton Stroke-type Transformer for Skeleton-based Action Recognition in Racket Sports", "url": "https://arxiv.org/html/2502.21085v2"}]

I found the paper! The arXiv paper 2308.12199 is titled "Towards Real-Time Analysis of Broadcast Badminton Videos".

However, I should note that while I found the paper and its abstract in the search results, the specific phrase "rally is at least 3 seconds" does not appear in the search result snippets I retrieved. The paper does discuss defining a "rally" frame as a frame where the actual gameplay is happening and includes a point segmentation method that classifies between rally and non-rally frames, but the 3-second minimum duration criterion is not mentioned in the available excerpts.

If you need to find that specific detail about the 3-second threshold, you may want to access the full PDF of the paper directly from arXiv.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

8. assistant (2026-06-06T17:34:07.990Z)

The paper is confirmed: "Towards Real-Time Analysis of Broadcast Badminton Videos" by Nilesh, Sharma, Ghosh, Jawahar (badminton). Now I need to verify the exact quote and its context. The PDF is too large for WebFetch, but the HTML version on arxiv may work. Let me try the HTML version and search for the specific quote.

9. user (2026-06-06T17:34:10.420Z)

The server returned HTTP 404 Not Found.

The response body was not retrieved. If this URL requires authentication, use an authenticated tool (e.g. `gh` for GitHub, or an MCP-provided fetch tool) instead of WebFetch.

10. user (2026-06-06T17:34:22.640Z)

Web search results for query: "\"Towards Real-Time Analysis of Broadcast Badminton Videos\" rally segmentation \"3 seconds\" minimum gap post-processing"

Links: [{"title": "Towards Real-Time Analysis of Broadcast Badminton Videos", "url": "https://arxiv.org/pdf/2308.12199"}, {"title": "Badminton Video Analysis based on Spatiotemporal and Stroke Features", "url": "https://www.researchgate.net/publication/317159692_Badminton_Video_Analysis_based_on_Spatiotemporal_and_Stroke_Features"}, {"title": "[2308.12199] Towards Real-Time Analysis of Broadcast Badminton"}]

Videos", "url": "https://arxiv.org/abs/2308.12199"}, {"title": "Towards Structured Analysis of Broadcast Badminton Videos | Request PDF", "url": "https://www.researchgate.net/publication/325002967_Towards_Structured_Analysis_of_Broadcast_Badminton_Videos"}, {"title": "VideoBadminton: A Video Dataset for Badminton Action Recognition", "url": "https://arxiv.org/html/2403.12385v1"}, {"title": "TennisExpert: Towards Expert-Level Analytical Sports Video Understanding", "url": "https://arxiv.org/pdf/2603.13397"}, {"title": "BST: Badminton Stroke-type Transformer for Skeleton-based Action Recognition in Racket Sports", "url": "https://arxiv.org/html/2502.21085v2"}, {"title": "VideoBadminton: A Video Dataset for Badminton Action Recognition | Request PDF", "url": "https://www.researchgate.net/publication/388099044_VideoBadminton_A_Video_Dataset_for_Badminton_Action_Recognition"}]

Great! I found the paper. Now let me search for the specific information about the "3 seconds" minimum gap post-processing mentioned in your query.

Links: [{"title": "VideoBadminton: A Video Dataset for Badminton Action Recognition", "url": "https://arxiv.org/html/2403.12385v1"}, {"title": "Towards Real-Time Analysis of Broadcast Badminton Videos", "url": "https://arxiv.org/pdf/2308.12199"}, {"title": "MonoTrack: Shuttle trajectory reconstruction from monocular badminton video", "url": "https://arxiv.org/pdf/2204.01899"}, {"title": "Benchmarking Badminton Action Recognition with a New Fine-Grained Dataset", "url": "https://arxiv.org/html/2403.12385v2"}, {"title": "Bridging the Gap: Doubles Badminton Analysis with Singles-Trained Models", "url": "https://arxiv.org/html/2508.13507v1"}, {"title": "ShuttleSet: A Human-Annotated Stroke-Level Singles Dataset for Badminton Tactical Analysis", "url": "https://arxiv.org/pdf/2306.04948"}, {"title": "TennisExpert: Towards Expert-Level Analytical Sports Video Understanding", "url": "https://arxiv.org/pdf/2603.13397"}, {"title": "One-Shot Badminton Shuttle Detection for Mobile Robots", "url": "https://arxiv.org/html/2603.06691v2"}, {"title": "BFMD: A Full-Match Badminton Dense Dataset for Dense Shot Captioning", "url": "https://arxiv.org/pdf/2603.25533"}]

Let me search more specifically for the post-processing details with the 3-second minimum gap threshold in the "Towards Real-Time Analysis" paper.

Links: [{"title": "[2308.12199] Towards Real-Time Analysis of Broadcast Badminton Videos", "url": "https://arxiv.org/abs/2308.12199"}, {"title": "Towards Real-Time Analysis of Broadcast Badminton Videos", "url": "https://arxiv.org/pdf/2308.12199"}, {"title": "Towards Real-Time Analysis of Broadcast Badminton Videos | DeepAI", "url": "https://deepai.org/publication/towards-real-time-analysis-of-broadcast-badminton-videos"}, {"title": "Towards Real-Time Analysis of Broadcast Badminton Videos", "url": "https://ui.adsabs.harvard.edu/abs/arXiv:2308.12199"}, {"title": "An All Deep System for Badminton Game Analysis", "url": "https://arxiv.org/pdf/2308.12645"}, {"title": "Towards Structured Analysis of Broadcast Badminton Videos", "url": "https://arxiv.org/pdf/1712.08714"}, {"title": "Future Prediction of Shuttlecock Trajectory in Badminton Using Player?s Information", "url": "https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10219238/"}]

Based on my web search, I found the paper "Towards Real-Time Analysis of Broadcast Badminton Videos". The paper describes a system for analyzing broadcast badminton videos with a focus on player movement analysis.

Regarding rally segmentation, a "rally" frame is defined as a frame where the actual gameplay is happening and a "non-rally" frame as a frame where the replay (highlights) has been shown, or players are resting, etc. The system allows grouping rally frames into rallies according to the points scored by the players and discarding the non-rally frames.

However, I was unable to locate the specific information about the "3 seconds minimum gap" post-processing threshold you mentioned in your search query within the available search results. The detailed technical specifications about post-processing parameters may be in the full PDF of the paper. I recommend accessing the full paper directly from arXiv (arxiv.org/pdf/2308.12199) to find those specific implementation details.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

11. assistant (2026-06-06T17:34:25.611Z)

The HTML version returned 404. I need to access the actual PDF text to verify the exact quote and its context. Let me download the PDF directly and extract the relevant text, since WebFetch can't handle the file size.

12. user (2026-06-06T17:34:27.263Z)

```
-rw-r--r-- 1 avidu 197609 11475365 Jun  6 23:04 2308.12199.pdf
2308.12199.pdf: PDF document, version 1.5
```

13. assistant (2026-06-06T17:34:29.339Z)

Downloaded the 11MB PDF. Now let me extract the text and search for the rally constraint quote.

14. user (2026-06-06T17:34:32.510Z)

PAGES 8 CHARS 34798

15. assistant (2026-06-06T17:34:35.195Z)

Text extracted (8 pages). Now let me search for the specific quote about "3 seconds" and rally constraints.

16. user (2026-06-06T17:34:36.016Z)

```
<tool_use_error>Path does not exist: /tmp/paper.txt. Note: your current working directory is
C:\Users\avidu\Projects\badminton-highlight-indexer.</tool_use_error>
```

17. assistant (2026-06-06T17:34:39.570Z)

The /tmp path resolved differently. Let me find the actual location of the file.

18. user (2026-06-06T17:34:41.195Z)

```
TEMP=C:\Users\avidu\AppData\Local\Temp
-rw-r--r-- 1 avidu 197609 35684 Jun  6 23:04 C:\Users\avidu\AppData\Local\Temp/paper.txt
-rw-r--r-- 1 avidu 197609 35684 Jun  6 23:04 /tmp/paper.txt
TMPDIR C:\Users\avidu\AppData\Local\Temp
```

19. user (2026-06-06T17:34:44.784Z)

```
65-the ?top-player? and player in yellow bounding box refers to
66-the ?bottom-player?. Dot at the center-bottom of the player?s
67:bounding box represents their current location. The ?Rally
68:Duration? is the time length of a particular rally. Four such
69-rallies are depicted for reference. (best viewed in color)
70-wins, unforced errors, successful smashes, distance covered,
--
83-complex human pose and motion makes real-time analysis
84-a complex task. Moreover, broadcast delay for a typical
85:badminton match is less than 15 seconds, usually 7 seconds
86-arXiv:2308.12199v1 [cs.CV] 23 Aug 2023
87-2. Many of the existing analysis methods are constrained by
--
115-25 rallies badminton match video of our own recording
116-having the ground truth data for the distance covered by
117:the players in each rally.
118-II. RELATED WORK
119-Sports analysis and Player Localization: Traditionally, work
--
224-A. Point Segmentation Annotation
225-Broadcast badminton matches are divided into gameplays
```

226:and replays. We define a ?rally? frame as a frame where the
227:actual gameplay is happening and a ?non-rally? frame as a
228:frame where the replay (highlights) has been shown, or players
229:are resting, etc. (or in other words whether the gameplay is
230:not happening). Generally, non-rally frames contain redundant
231:information about the game and do not contribute much in
232:the analysis part. Therefore, we first start classifying between
233:rally and non-rally frames. We call this classification method
234:as Point Segmentation . This allows us to keep all the rally
235:frames, group them into rallies according to the points scored
236:by the players and discard the non-rally frames. To make the
237:dataset usable for point segmentation, we annotate every frame
238:of a match as a rally or non-rally frame using the ELAN video
239:annotation tool 4. We annotate 23156 frames in total, where
240:15147 were non-rally frames and 8009 were rally frames.

241-B. Player Bounding Boxes Annotation

242-We annotate the dataset to get the bounding box for each

--

252-corresponds to the player on the near side of the court with
253-respect to the viewpoint of the camera. We randomly pick
254:150 rally frames from each match and annotated them for
255:player bounding boxes for both the players. Thus, we end up
256-with 3000 (150 frames $\times$ 2 players $\times$ 10 matches) annotated

--

276-analysis.

277-which were accurate up to 1 meter, measured the distance
278:covered by each player during each rally. We recorded 25
279-rallies and obtained the distance values for each player for
280-all of them. All the videos were recorded at approximately

--

290-can be used for real-time analysis of videos.

291-A. Point Segmentation

292:The very first step of our pipeline is to keep the rally
293:frames and discard the non-rally frames (which are redundant
294:for our analysis) from the given match video. Rally frames
295:are usually always shot from the broadcast camera view
296:whereas the non-rally frames can include close-ups of players,
297:additional graphics (score cards) and unusual camera angles.
298-These differences help distinguish these frames. Unlike Ghosh
299-et al [4] who use HOG + SVM, we use a convolutional
300:neural network based classifier to classify rally and non-rally
301:frames. We finetune the ResNet-18 [24] model (pre-trained
302:on ImageNet [25] dataset) by changing the last layer of this
303:network to label the frame as rally frame or non-rally frame.
304-As shown in table II, our CNN based method is both more
305:accurate and also exhibits lower latency compared to HOG +
306-SVM.

307-At test time, we classify the frames retrieved from the match
308:video and number the rally frames in a sequential manner.
309-We assume that a rally is at least 3 seconds long and there
310:is a minimum of 3 seconds gap between two rallies. This
311-allows us to group frames of the match into individual rallies
312:by separating sets of continuous rally frames with at least
313:3 seconds of non-rally frames in between them. Thus Point
314-Segmentation gives us a set of rallies from the match. We
315:number each rally sequentially as rally-1, rally-2 etc.

316-B. Player Localization

317-Once we obtain individual rallies, we need to find player
318:location in each frame of each rally. A YOLOv3 [26] network
319-is finetuned for these two classes. In order to overcome the
320:constraint of only having 3000 annotated samples for player

--

351-mAP@0.5

352-Player Bottom

353-Rally Latency

354:(Seconds)
355-Faster R-CNN 96.90 97.85 76.63
356-YOLOv3 98.07 98.13 0.14
--
381-player's position and footwork information around the court
382-for a particular gameplay. This view can also be seen as
383:heatmap of the players for that particular rally. (Best viewed
384-in color)
385-D. Analysis
--
390-and their coaches to analyze the game.
391-Player Distance Calculation: To compute the distance
392:covered by the player in a particular rally, we utilize our player
393:tracks in top view space. Now the player locations are in 2-
394-dimensional space and we take the center bottom point of
--
408-unitary method to scale this pixel-based distance into meters to
409-get the actual distance values. We add all the distances (meters)
410:successively for the whole rally to calculate the distance for
411:each player in that particular rally. The distance calculation of
412-6<http://www.worldbadminton.com/rules/>
413:a rally for one player is formulated as:
414- $n \times 1X$
415- $i=1$
--
419- $\sum_{i=1}^n (y_{i+1} - y_i)^2 \times 13.41$
420-Dv
421:where n is the number of frames in the rally, xi and yi are
422-the co-ordinates of the player in the top-view space at ith
423-frame, Dh is the pixel-based distance from top-left corner
--
433-calculate the distance.
434-Player Average Speed Calculation: To calculate the average
435:speed achieved by each of the players in a particular rally,
436-we use the distance covered by the players. As we know the
437:number of frames in a particular rally, we find out the total
438:time elapsed (25 frames per second) and divide the player
439:distance covered by the total time.
440-V. EVALUATIONS
--
443-A. Player Distance Calculation Evaluation
444-To evaluate the accuracy of our proposed pipeline in terms
445:of the distance covered by the players in a particular rally, we
446-used our recorded badminton match video which has ground
447-truth data of 25 rallies for the distance covered by the players.
448-We use our proposed pipeline to obtain predicted distance
449:values for each rally and compare them to the ground truth
450-values. We used the same evaluation metric as [29], which
451-uses paired t-tests to calculate differences between the ground
452-truth distances and predicted distances. We report the mean,
453:standard deviation, minimum, and maximum error between the
454-ground truth and predicted values for both Player Top and
455-Player Bottom. We also report the t-stats value and degree
--
463-tance and the predicted distance covered by the players in our
464-own recorded videos. It should be noted that maximum error
465:we got is 6 meters and minimum error is 1 meter for player
466-top and player bottom respectively.
467-Fig. 6: Comparison between time taken for gameplay and time
--
478-same scenario is not feasible, we use a simulation of the real-
479-time scenario to evaluate the performance (with respect to
480:latency) of our model. Generally, as live matches are streamed,
481:the camera recording the match captures 25 frames per second
482-and thus the match video file size increases with time. In order

483-to simulate the real-time scenario, we fed the pre-recorded
484-match videos to our proposed pipeline at precisely 25 frames
485:per second, sequentially. Thus our models did not have access
486-to all the frames at the same time. We ran this simulation on
487-61 rallies (totaling up to 55 minutes) and timed our models.

--

493-for our experiments. We attempt to extract the analysis from
494-the data we got using the methods mentioned above. We have
495:segmented the frames into either rally or non-rally frame using
496:point segmentation method which in the end, gives us rally
497:directories as soon as the rally is over in the live match, where
498:each directory contains rally frames for that specific rally. Then
499:we pass the rally frames to the player localization method to
500-get the player bounding boxes. We perform homography to
501-change the space from broadcast camera view to top-view.
502-We perform each analysis on frame level for the accessed file.
503-7<http://www.pbl-india.com/>
504:On an average, one rally is 10 to 12 seconds long and it con-
505:tains 250 to 300 rally frames. For a real time implementation, it
506-is not feasible to process these many frames in one go. Hence,
507-to solve this problem for this scenario, we process every 3rd
508:rally frame for player localization method. As there are 25
509:frames per second in the broadcast match video, we assume
510-that there is no significant change in player location if we
511:take every third rally frame. After getting the rally directories
512:from the above point segmentation method we pass the rally
513:directories to the player localization method to get the player
514-location for each of the frames in a parallel manner. Using
515:this rally-level parallelism for analysis helped us to achieve
516-the goal in the real-time scenario.

517-VII. FIELD TRIAL

--

555-camera from the approximately same broadcast angle as the
556-other datasets. This dataset also consists of ground truth data
557:for the distance covered by the players in each rally (which
558-was obtained from the distance trackers worn by the players
559-during gameplay). We hope that access to these datasets helps

20. assistant (2026-06-06T17:35:00.412Z)

I have found the exact passage. The relevant text appears at lines 307-315 in the "Point Segmentation" section of the methodology.

The exact text reads:

> "At test time, we classify the frames retrieved from the match video and number the rally frames in a sequential manner. **We assume that a rally is at least 3 seconds long and there is a minimum of 3 seconds gap between two rallies.** This allows us to group frames of the match into individual rallies by separating sets of continuous rally frames with at least 3 seconds of non-rally frames in between them. Thus Point Segmentation gives us a set of rallies from the match."

Now let me evaluate the claim against the checklist. The claim states the method "imposes the same heuristic temporal constraints the user's pipeline uses (minimum rally duration and minimum inter-rally gap): a rally is at least 3 seconds long with a minimum 3-second gap between rallies."

Let me assess each element:

1. **Quote support**: The quote is verbatim from the paper. The paper explicitly states this is an assumption used by their method at test time to group frames into rallies ? i.e., a post-processing heuristic on top of the CNN frame classifier. This is exactly a "minimum rally duration AND minimum inter-rally gap" constraint. Supported.

2. **Contradicting evidence**: None found. The phrase is unique to this paper and used as

described.

3. ****Source quality****: This is the primary source (the arxiv paper itself, peer-context IEEE/conference-style paper by IIIT Hyderabad's CVIT group ? Jawahar et al). For a factual claim about what the paper says, the primary source is the gold standard. Sufficient.

4. ****Outdated****: The claim is about what this 2023 paper states; it's an accurate citation of a static document. Not an issue.

5. ****Marketing/cherry-pick****: No ? it's a methodological detail in a research paper, not a benchmark or marketing claim.

One nuance to check for over-reach: the claim says these are "the same heuristic temporal constraints the user's pipeline uses." The user's pipeline (per context) uses velocity/merge-gap/min-duration thresholds. The paper's "min rally duration" + "min inter-rally gap" maps conceptually onto the user's "min-duration" and "merge-gap" thresholds. The 3s/3s values are precisely as quoted. The framing is fair ? the paper does impose these as heuristic post-processing constraints, parallel in kind to the user's. This is a faithful description, not an overreach.

The claim is well-supported by the verbatim quote, drawn from the primary source, current, and not marketing. The claim is NOT refuted.

21. user (2026-06-06T17:35:07.743Z)

Structured output provided successfully

22. assistant (2026-06-06T17:35:16.881Z)

Verification complete. The claim is NOT refuted ? it is well-supported by the verbatim quote from the primary source (arXiv 2308.12199, Section IV.A "Point Segmentation"), which I confirmed by downloading and extracting the full PDF. The paper explicitly imposes both a minimum 3-second rally duration and a minimum 3-second inter-rally gap as a test-time heuristic to group classified rally frames into discrete rallies, faithfully matching the claim's description.